Advanced Electromagnetics

Chapter 1 – Maxwell's Equations

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Outline

1.1 Formulation of Maxwell's Equations
- 1.1.1 Differential and Integral Form
- 1.1.2 Duality Theorem

1.2 Constitutive Relations

1.3 Boundary Conditions

1.4 Wave Equations

1.5 Time Harmonic Form

1.6 Poynting Theorem
- 1.6.1 Time-Domain Formulation
- 1.6.2 Frequency-Domain Formulation

1.7 Uniqueness Theorem

1.1 Formulation of Maxwell's Equations

1.1.1 Differential and Integral Form

Faraday's law

$$\vec{\nabla} \times \vec{E} = -\partial_t \vec{B} - \vec{M} \tag{1.1.1}$$

Maxwell-Ampere's Law

$$\vec{\nabla} \times \vec{H} = \partial_t \vec{D} + \vec{J} \tag{1.1.2}$$

Continuity relation

$$\vec{\nabla} \cdot \vec{J} = -\partial_t \rho, \quad \vec{\nabla} \cdot \vec{M} = -\partial_t \varrho \tag{1.1.3}$$

*Lorentz equation of force

$$\vec{F} = q\left(\vec{E} + \vec{v} \times \vec{B}\right) \tag{1.1.4}$$

1.1.1 Differential and Integral Form

- $\vec{E}$: electric field intensity (V/m)
- $\vec{H}$: magnetic field intensity (A/m)
- $\vec{D}$: electric flux density (C/m^2)
- $\vec{B}$: magnetic flux density (T)
- $\vec{J}$: volumetric electric current density (A/m^2)
- ρ: electric charge density (C/m^3)
- $\vec{M}$: volumetric magnetic current density (V/m^2)
- ϱ: magnetic charge density (Wb/m^3)

1.1.1 Differential and Integral Form

Taking divergence of (1.1.2)

$$\vec{\nabla} \cdot (\vec{\nabla} \times \vec{H}) = \partial_t \vec{\nabla} \cdot \vec{D} + \vec{\nabla} \cdot \vec{J} \equiv 0 \qquad (1.1.5)$$

And from (1.1.3) we get

$$\partial_t (\vec{\nabla} \cdot \vec{D} - \rho) \equiv 0 \qquad (1.1.6)$$

This implies that

$$\vec{\nabla} \cdot \vec{D} - \rho = C(x, y, z) \qquad (1.1.7)$$

similarly

$$\vec{\nabla} \cdot \vec{B} - \varrho = C'(x, y, z) \qquad (1.1.8)$$

1.1.1 Differential and Integral Form

In (1.1.7), if $C \neq 0$, it can be absorbed into ρ. The case is similar to (1.1.8). Thus, we can set $C = C' = 0$.

We get the Gauss's law

$$\vec{\nabla} \cdot \vec{D} = \rho \tag{1.1.10}$$

$$\vec{\nabla} \cdot \vec{B} = \varrho \tag{1.1.11}$$

Note that (1.1.10) and (1.1.11) are not independent of (1.1.1)-(1.1.3).

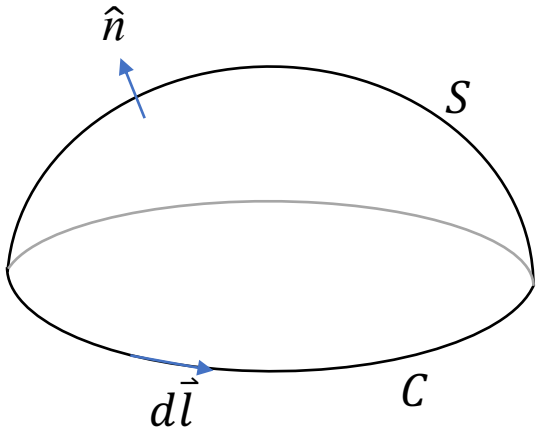
1.1.1 Differential and Integral Form

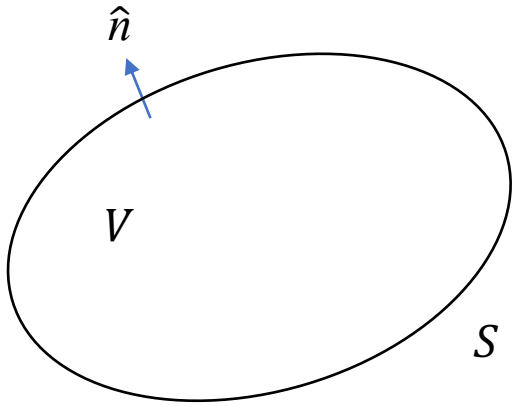
Stokes' theorem

$$\int_S \vec{\nabla} \times \vec{\Psi} \cdot d\vec{s} = \oint_C \vec{\Psi} \cdot d\vec{l} \qquad (1.1.12)$$

Divergence theorem

$$\int_V \vec{\nabla} \cdot \vec{\Psi}\, dv = \oint_S \vec{\Psi} \cdot d\vec{s} \qquad (1.1.13)$$





$d\vec{s} = \hat{n} ds$

1.1.1 Differential and Integral Form

Applying (1.1.12) to (1.1.1) and (1.1.2), we get

$$\oint_C \vec{E} \cdot d\vec{l} = -\int_S (\partial_t \vec{B} + \vec{M}) \cdot d\vec{s} \tag{1.1.14}$$

$$\oint_C \vec{H} \cdot d\vec{l} = \int_S (\partial_t \vec{D} + \vec{J}\,) \cdot d\vec{s} \tag{1.1.15}$$

Applying (1.1.13) to (1.1.3), (1.1.10) and (1.1.11), we get

$$\oint_S \vec{J} \cdot d\vec{s} = -\partial_t \int_V \rho \, dv \tag{1.1.16}$$

$$\oint_S \vec{D} \cdot d\vec{s} = \int_V \rho \, dv \tag{1.1.17}$$

$$\oint_S \vec{B} \cdot d\vec{s} = \int_V \varrho \, dv \tag{1.1.18}$$

1.1.1 Differential and Integral Form

Another theorem for the curl operator

$$\int_V \vec{\nabla} \times \vec{\Psi}\, dv = \oint_S \left(\hat{n} \times \vec{\Psi}\right) \cdot d\vec{s} \qquad (1.1.19)$$

Applying (1.1.19) to (1.1.1) and (1.1.2), we get

$$\oint_S \hat{n} \times \vec{E} \cdot d\vec{s} = -\int_V \left(\partial_t \vec{B} + \vec{M}\right) dv \qquad (1.1.20)$$

$$\oint_S \hat{n} \times \vec{H} \cdot d\vec{s} = \int_V \left(\partial_t \vec{D} + \vec{J}\right) dv \qquad (1.1.21)$$

1.1.2 Duality Theorem

It is noted that when fictitious magnetic sources ϱ and $\vec{M}$ are introduced, Maxwell's equations become symmetric. By exploiting symmetry, Maxwell's equations can be separated into one set involving only electric sources and another involving only magnetic sources.

$$\begin{cases} \vec{\nabla} \times \vec{E} = -\mu \partial_t \vec{H} \\ \vec{\nabla} \times \vec{H} = \epsilon \partial_t \vec{E} - \vec{J} \\ \vec{\nabla} \cdot \epsilon \vec{E} = \rho \\ \vec{\nabla} \cdot \mu \vec{H} = 0 \end{cases} \quad \& \quad \begin{cases} \vec{\nabla} \times \vec{E} = -\mu \partial_t \vec{H} - \vec{M} \\ \vec{\nabla} \times \vec{H} = \epsilon \partial_t \vec{E} \\ \vec{\nabla} \cdot \epsilon \vec{E} = 0 \\ \vec{\nabla} \cdot \mu \vec{H} = \varrho \end{cases}$$

(1.1.22a) & (1.1.22b)

1.1.2 Duality Theorem

When the following transformations are applied, equations (1.1.22a) and (1.1.22b) are interchanged:

$$\begin{aligned} \vec{E} &\to \vec{H}', & \vec{H} &\to -\vec{E}' \\ \vec{J} &\to \vec{M}', & \vec{M} &\to -\vec{J}' \\ \rho &\to \varrho', & \varrho &\to -\rho' \\ \epsilon &\to \mu', & \mu &\to \epsilon' \end{aligned} \tag{1.1.23}$$

When applying duality to same medium ($\eta = \sqrt{\mu/\epsilon}$)

$$\begin{aligned} \vec{E} &\to \eta\vec{H}', & \vec{H} &\to -\vec{E}'/\eta \\ \vec{J} &\to \vec{M}'/\eta, & \vec{M} &\to -\eta\vec{J}' \\ \rho &\to \varrho'/\eta, & \varrho &\to -\eta\rho' \end{aligned} \tag{1.1.24}$$

1.2 Constitutive Relations

1.2 Constitutive Relations

Excluding $\vec{M}$ and ϱ (non-physical), there are 5 vectors and 1 scalar, resulting in 16 unknowns.

From (1.1.1)-(1.1.3), there are 7 scalar equations, which means extra 9 equations are needed to make the system determinate.

The constitutive relations relates $\vec{D}, \vec{B}, \vec{J}$ with $\vec{E}, \vec{H}$ by

$$\begin{cases} \vec{D} = \bar{\bar{C}}_1(\vec{E}, \partial_t \vec{E}, \partial_t^2 \vec{E}, \dots \vec{H}, \partial_t \vec{H}, \partial_t^2 \vec{H}, \dots) \\ \vec{H} = \bar{\bar{C}}_2(\vec{E}, \partial_t \vec{E}, \partial_t^2 \vec{E}, \dots \vec{H}, \partial_t \vec{H}, \partial_t^2 \vec{H}, \dots) \\ \vec{J} = \bar{\bar{C}}_3(\vec{E}, \partial_t \vec{E}, \partial_t^2 \vec{E}, \dots \vec{H}, \partial_t \vec{H}, \partial_t^2 \vec{H}, \dots) \end{cases} \tag{1.2.1}$$

In general, $\bar{\bar{C}}_j$ $(j = 1, 2, 3)$ are tensor functions of time.

1.2 Constitutive Relations

We confine our exposition on the material properties by the following restrictions :

- Stationary: $\bar{\bar{C}}_j$ are not functions of time
- Non-chiral: $\bar{\bar{C}}_1, \bar{\bar{C}}_3$ are related to $\vec{E}$ and $\bar{\bar{C}}_2$ is related to $\vec{H}$
- Linear: $\bar{\bar{C}}_j$ are related to $\vec{E}$ and $\vec{H}$ only (no higher derivatives)

Often simplifications can be made if the medium is

- Isotropic: $\bar{\bar{C}}_j$ are scalars, i.e., C_j (otherwise they are called anisotropic)
- Homogeneous: $\bar{\bar{C}}_j$ are not functions of space
- In this note, we call a medium simple if it is linear, isotropic and homogeneous.

1.2 Constitutive Relations

When a linear dielectric medium is perturbed by an electric field, the constitutive relation for $\vec{D}$ and $\vec{E}$ is

$$\vec{D} = \epsilon_0 \vec{E} + \vec{\mathcal{P}} \tag{1.2.2}$$

ϵ_0 is the electric permittivity of the vacuum (8.854×10^{-12} F/m), and $\vec{\mathcal{P}}$ is the electric polarization defined as

$$\vec{\mathcal{P}} = \bar{\bar{\chi}}_e \epsilon_0 \vec{E} \tag{1.2.3}$$

$\bar{\bar{\chi}}_e$ is the electric susceptibility tensor. If the medium is isotropic, then $\bar{\bar{\chi}}_e$ becomes a scalar, and $\vec{\mathcal{P}}$ is parallel to $\vec{E}$. We can define

$$\vec{D} = \bar{\bar{\epsilon}} \vec{E} \tag{1.2.4}$$

$$\bar{\bar{\epsilon}} = \epsilon_0 (\bar{\bar{I}} + \bar{\bar{\chi}}_e) \tag{1.2.5}$$

1.2 Constitutive Relations

Similarly in magnetic medium, we have

$$\vec{B} = \mu_0 \vec{H} + \vec{\mathcal{M}} \tag{1.2.6}$$

μ_0 is the magnetic permeability of the vacuum ($4\pi \times 10^{-7}$ H/m), and $\vec{\mathcal{M}}$ is the magnetic polarization defined as

$$\vec{\mathcal{M}} = \bar{\bar{\chi}}_m \mu_0 \vec{H} \tag{1.2.7}$$

$\bar{\bar{\chi}}_m$ is the magnetic susceptibility tensor. We can define

$$\vec{B} = \bar{\bar{\mu}} \vec{H} \tag{1.2.8}$$

$$\bar{\bar{\mu}} = \mu_0 (\bar{\bar{I}} + \bar{\bar{\chi}}_m) \tag{1.2.9}$$

1.2 Constitutive Relations

The constitutive relation between $\vec{J}$ and $\vec{E}$ is

$$\vec{J} = \bar{\bar{\sigma}}\,\vec{E} \tag{1.2.10}$$

$\bar{\bar{\sigma}}$ is the conductivity of the medium. The conductivity in vacuum is 0 (S/m).

For a simple medium, we can simplify the tensors $\bar{\bar{\epsilon}}$, $\bar{\bar{\mu}}$ and $\bar{\bar{\sigma}}$ to ϵ, μ and σ, respectively. (1.1.1) and (1.1.2) can be rewritten as

$$\vec{\nabla} \times \vec{E} = -\mu \partial_t \vec{H} \tag{1.2.11}$$

$$\vec{\nabla} \times \vec{H} = (\sigma + \epsilon \partial_t)\vec{E} \tag{1.2.12}$$

1.3 Boundary Conditions

1.3 Boundary Conditions

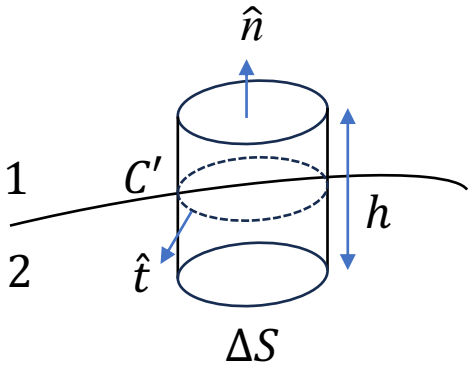
To derive boundary conditions, integral form of Maxwell's equations are needed.

Applying (1.1.20) to the pillbox

$$\hat{n} \times \left(\vec{E}_1 - \vec{E}_2\right)\Delta S + \int_{side} \hat{t} \times \vec{E} \cdot d\vec{s} = -\Delta S \int_{-h/2}^{h/2} (\partial_t \vec{B} + \vec{M}) d\zeta \quad (1.3.1)$$

when $h \to 0$, we define the surface magnetic current density (V/m) as

$$\vec{M}_s = \lim_{h \to 0} \int_{-h/2}^{h/2} \vec{M} d\zeta \quad (1.3.2)$$



1.3 Boundary Conditions

we get

$$\hat{n} \times \left(\vec{E}_1 - \vec{E}_2\right) = -\vec{M}_s \tag{1.3.3}$$

Similarly, applying (1.1.21) to the pillbox, when $h \to 0$, we define the surface electric current density (A/m) as

$$\vec{J}_s = \lim_{h \to 0} \int_{-h/2}^{h/2} \vec{J} d\zeta \tag{1.3.4}$$

we get

$$\hat{n} \times \left(\vec{H}_1 - \vec{H}_2\right) = \vec{J}_s \tag{1.3.5}$$

Note that $\vec{J}_s$ exists only when one of the medium's $\sigma \to \infty$.

1.3 Boundary Conditions

Applying (1.1.17) to the pillbox

$$\hat{n} \cdot \left(\vec{D}_1 - \vec{D}_2\right)\Delta S + \int_{side} \hat{t} \cdot \vec{D}\, ds = \Delta S \int_{-h/2}^{h/2} \rho d\zeta \tag{1.3.6}$$

when $h \to 0$, we define the surface charge density (C/m^2) as

$$\rho_s = \lim_{h \to 0} \int_{-h/2}^{h/2} \rho d\zeta \tag{1.3.7}$$

we get

$$\hat{n} \cdot \left(\vec{D}_1 - \vec{D}_2\right) = \rho_s \tag{1.3.8}$$

Similarly, from (1.1.18) we get

$$\hat{n} \cdot \left(\vec{B}_1 - \vec{B}_2\right) = \varrho_s \tag{1.3.9}$$

1.3 Boundary Conditions

Lastly, applying (1.1.16) to the pillbox

$$\hat{n} \cdot \left(\vec{J}_1 - \vec{J}_2\right)\Delta S + \int_{side} \hat{t} \cdot \vec{J}\, ds = -\Delta S \int_{-h/2}^{h/2} \partial_t \rho d\zeta \qquad (1.3.10)$$

when $h \to 0$, we can express the side integral as

$$\oint_{C'} \lim_{h\to 0} \int_{-h/2}^{h/2} \vec{J} d\zeta \cdot \hat{t} dl = \oint_{C'} \vec{J}_s \cdot \hat{t} dl = \int_{S'} \vec{\nabla}_s \cdot \vec{J}_s\, ds \qquad (1.3.11)$$

we get

$$\left(\vec{J}_1 - \vec{J}_2\right) + \vec{\nabla}_s \cdot \vec{J}_s = -\partial_t \rho_s \qquad (1.3.12)$$

Similarly

$$\left(\vec{M}_1 - \vec{M}_2\right) + \vec{\nabla}_s \cdot \vec{M}_s = -\partial_t \rho_s \qquad (1.3.13)$$

1.4 Wave Equations

1.4 Wave Equations

Consider a lossless simple medium, taking the curl of (1.1.1)

$$\vec{\nabla} \times \vec{\nabla} \times \vec{E} = \vec{\nabla}\left(\vec{\nabla} \cdot \vec{E}\right) - \nabla^2 \vec{E} = -\partial_t\left(\vec{\nabla} \times \mu \vec{H}\right) \qquad (1.4.1)$$

$$\nabla^2 \vec{E} - \mu\epsilon\partial_t^2 \vec{E} = \mu\partial_t \vec{J} + \vec{\nabla}(\rho/\epsilon) + \vec{\nabla} \times \vec{M} \qquad (1.4.2)$$

(1.4.2) is the wave equation for electric field. We can obtain the wave equation for magnetic field by applying the duality transform (1.1.23) to (1.4.2)

$$\nabla^2 \vec{H} - \mu\epsilon\partial_t^2 \vec{H} = -\vec{\nabla} \times \vec{J} + \vec{\nabla}(\varrho/\mu) + \epsilon\partial_t \vec{M} \qquad (1.4.3)$$

1.4 Wave Equations

(1.4.2) and (1.4.3) form a set of coupled inhomogeneous DE

$$\Box \begin{pmatrix} \vec{E} \\ \vec{H} \end{pmatrix} = \begin{pmatrix} \mu \partial_t \vec{J} + \vec{\nabla}\left(\frac{\rho}{\epsilon}\right) + \vec{\nabla} \times \vec{M} \\ -\vec{\nabla} \times \vec{J} + \vec{\nabla}\left(\frac{\varrho}{\mu}\right) + \epsilon \partial_t \vec{M} \end{pmatrix} \tag{1.4.4}$$

$\Box \equiv \nabla^2 - v^{-2}\partial_t^2$ is the d'Alembert operator with $v = (\mu\epsilon)^{-1/2}$.

1.5 Time Harmonic Form

1.5 Time Harmonic Form

A time domain signal can be decomposed into a spectrum of time harmonic components

$$f(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} g(\omega)e^{i\omega t}d\omega \qquad (1.5.1)$$

with

$$g(\omega) = \int_{-\infty}^{\infty} f(t)e^{-i\omega t}d\omega \qquad (1.5.2)$$

The Fourier and inverse Fourier transform relationship is

$$g(\omega) = \mathfrak{F}[f(t)]$$

$$f(t) = \mathfrak{F}^{-1}[g(\omega)] \qquad (1.5.3)$$

1.5 Time Harmonic Form

Now we consider the harmonic electric field at angular frequency ω. We consider its phasor form

$$\vec{E}(\vec{r},t) = \Re\left[\vec{E}(\vec{r},\omega)e^{i\omega t}\right] \tag{1.5.4}$$

Note that we use the same notation for both the time domain field and frequency domain field, and we will suppress its dependency when there is no ambiguity.

In general, the phasor field $\vec{E}(\vec{r},\omega)$ is a complex number.

1.5 Time Harmonic Form

The time harmonic Maxwell's equation takes the form

$$\vec{\nabla} \times \vec{E} = -i\omega\vec{B} - \vec{M} \tag{1.5.5}$$

$$\vec{\nabla} \times \vec{H} = i\omega\vec{D} + \vec{J} \tag{1.5.6}$$

$$\vec{\nabla} \cdot \vec{D} = \rho \tag{1.5.7}$$

$$\vec{\nabla} \cdot \vec{B} = \varrho \tag{1.5.8}$$

The method to translate the time-domain equation to frequency domain is to replace ∂_t with $i\omega$, and vice versa.

1.5 Time Harmonic Form

Analyzing (1.5.6), with $\vec{D} = \epsilon\vec{E}$ and $\vec{J} = \vec{J}_c + \vec{J}_i = \sigma\vec{E} + \vec{J}_i$, we get

$$\vec{\nabla} \times \vec{H} = i\omega\left(\epsilon - i\frac{\sigma}{\omega}\right)\vec{E} + \vec{J}_i \tag{1.5.9}$$

We define the complex permittivity as

$$\epsilon_c = \epsilon - i\frac{\sigma}{\omega} \tag{1.5.10}$$

In general $\epsilon = \epsilon' - i\epsilon''$. We define the loss tangent of the medium as

$$\tan\delta = \frac{\epsilon''}{\epsilon'} + \frac{\sigma}{\omega\epsilon'} \tag{1.5.11}$$

Medium with $\tan\delta \ll 1$ is characterized as good dielectric, and with $\tan\delta \gg 1$ is characterized as good conductor.

1.5 Time Harmonic Form

The d'Alembert operator in (1.4.4) in frequency domain becomes

$$\nabla^2 - \frac{(i\omega)^2}{1/\mu\epsilon_c} = \nabla^2 + k^2 \tag{1.5.12}$$

where

$$k = \omega\sqrt{\mu\epsilon_c} = k_R + ik_I \tag{1.5.13}$$

is the wavenumber. The square root of ϵ_c is chosen so that k_I relates to the physical attenuation of the wave propagation.

Several literatures uses $\gamma^2 = (\alpha + i\beta)^2 = -\omega^2\mu\varepsilon_c = -k^2$. The relationship between the real and imaginary parts are $\alpha = -k_I$ and $\beta = k_R$.

1.6 Poynting Theorem

1.6.1 Time-Domain Formulation

Considering the identity

$$\vec{\nabla} \cdot \left(\vec{E} \times \vec{H}\right) \equiv \vec{H} \cdot \vec{\nabla} \times \vec{E} - \vec{E} \cdot \vec{\nabla} \times \vec{H} \qquad (1.6.1)$$

We define the instantaneous Poynting vector $\vec{S} = \vec{E} \times \vec{H}$ (W/m^2)

$$\vec{\nabla} \cdot \vec{S} = \vec{H} \cdot \left(-\partial_t \vec{B} - \vec{M}\right) - \vec{E} \cdot \left(\partial_t \vec{D} + \vec{J}\right) =$$

$$-\partial_t \left(\frac{\mu}{2} \vec{H} \cdot \vec{H}\right) - \partial_t \left(\frac{\epsilon}{2} \vec{E} \cdot \vec{E}\right) - \vec{H} \cdot \vec{M} - \vec{E} \cdot \vec{J} \qquad (1.6.2)$$

1.6.1 Time-Domain Formulation

The term $\frac{\epsilon}{2}\vec{E}\cdot\vec{E} = w_e$ and $\frac{\mu}{2}\vec{H}\cdot\vec{H} = w_m$ denote the electric and magnetic energy density, and $\vec{H}\cdot\vec{M} + \vec{E}\cdot\vec{J} = p_l$ denotes the power loss/supply per unit volume. We rewrite the Poynting theorem as

$$\vec{\nabla}\cdot\vec{S} = -\partial_t(w_e + w_m) - p_l \tag{1.6.3}$$

and conduct integration over a finite volume V by applying the divergence theorem:

$$\oint_S \hat{n}\cdot\vec{S}ds = -\partial_t \int_V (w_e + w_m)dv - \int_V p_l dv \tag{1.6.4}$$

1.6.2 Frequency-Domain Formulation

From $\vec{E}(\vec{r},t) = \Re\left[\vec{E}(\vec{r})e^{i\omega t}\right] = \left(\vec{E}(\vec{r})e^{i\omega t} + \vec{E}^*(\vec{r})e^{-i\omega t}\right)/2$, and

$$\vec{S}(\vec{r},t) = \left(\Re\left[\vec{E}(\vec{r}) \times \vec{H}^*(\vec{r})\right] + \Re\left[\vec{E}(\vec{r}) \times \vec{H}(\vec{r})e^{i2\omega t}\right]\right)/2 \quad (1.6.5)$$

we have

$$\vec{S}_{\text{av}} = \frac{1}{T}\int_0^T \vec{E}(\vec{r},t) \times \vec{H}(\vec{r},t)dt = \frac{1}{2}\Re\left[\vec{E}(\vec{r}) \times \vec{H}^*(\vec{r})\right] \quad (1.6.6)$$

Define the vector

$$\vec{S}(\vec{r}) = \frac{1}{2}\vec{E} \times \vec{H}^* \quad (1.6.7)$$

as the complex Poynting vector.

1.6.2 Frequency-Domain Formulation

From the identity (1.6.1)

$$\vec{\nabla}\cdot\vec{S}=\vec{\nabla}\cdot\left(\frac{\vec{E}\times\vec{H}^*}{2}\right)=-i2\omega\left(\frac{\mu\left|\vec{H}\right|^2}{4}-\frac{\epsilon\left|\vec{E}\right|^2}{4}\right)-\frac{\vec{E}\cdot\vec{J}^*}{2}-\frac{\vec{H}^*\cdot\vec{M}}{2}=$$

$$-i2\omega(w_m-w_e)-p_l \qquad (1.6.8)$$

is the complex Poynting theorem, where $w_m=\mu\left|\vec{H}\right|^2/4$, $w_e=\epsilon\left|\vec{E}\right|^2/4$ and $p_l=\left(\vec{E}\cdot\vec{J}^*+\vec{H}^*\cdot\vec{M}\right)/2$ are the time average electric and magnetic energy density and the time average power loss/supply.

1.7 Uniqueness Theorem

1.7 Uniqueness Theorem

In a volume V, there exists sources $\vec{J}$ and $\vec{M}$. Assume the medium is simple but may be lossy, i.e, (ε, μ) are in gerneral complex. Our goal is to determine the conditions such that the electromagnetic field inside the volume V is unique.

Suppose there are two sets of solutions exist in V, say, $(\vec{E}_1, \vec{H}_1)$ and $(\vec{E}_2, \vec{H}_2)$. We want to find the conditions for $\delta\vec{E} = \vec{E}_1 - \vec{E}_2 = 0$ and $\delta\vec{H} = \vec{H}_1 - \vec{H}_2 = 0$ inside V.

1.7 Uniqueness Theorem

From (1.5.5) and (1.5.6), we have

$$\begin{cases} \vec{\nabla} \times \vec{E}_1 = -i\omega\mu\vec{H}_1 - \vec{M} \\ \vec{\nabla} \times \vec{H}_1 = i\omega\varepsilon\vec{E}_1 + \vec{J} \end{cases}, \begin{cases} \vec{\nabla} \times \vec{E}_2 = -i\omega\mu\vec{H}_2 - \vec{M} \\ \vec{\nabla} \times \vec{H}_2 = i\omega\varepsilon\vec{E}_2 + \vec{J} \end{cases} \tag{1.7.1}$$

inside V. Substracting the two sets of equations, we get

$$\begin{cases} \vec{\nabla} \times \delta\vec{E} = -i\omega\mu\delta\vec{H} \\ \vec{\nabla} \times \delta\vec{H} = i\omega\varepsilon\delta\vec{E} \end{cases} \tag{1.7.2}$$

1.7 Uniqueness Theorem

Dot the first equation in (1.7.2) with $\delta\vec{H}^*$ and the second equation with $\delta\vec{E}^*$, and substract the two:

$$\vec{\nabla}\times\delta\vec{E}\cdot\delta\vec{H}^* - \vec{\nabla}\times\delta\vec{H}\cdot\delta\vec{E}^* = \vec{\nabla}\cdot\left(\delta\vec{E}\times\delta\vec{H}^*\right)$$

$$= i\omega\left(\mu\left|\delta\vec{H}\right|^2 - \varepsilon^*\left|\delta\vec{E}\right|^2\right) \qquad (1.7.3)$$

Conducting integration of (2.1.3) over V

$$\oint_S \delta\vec{E}\times\delta\vec{H}^*\cdot d\vec{s} = \int_V i\omega\left(\mu\left|\delta\vec{H}\right|^2 - \varepsilon^*\left|\delta\vec{E}\right|^2\right)dv \qquad (1.7.4)$$

1.7 Uniqueness Theorem

The key point is, if the surface integral in (1.7.4) equals to zero, so does the volume integral, which implies

$$\begin{cases} \int_V \left(\Re[\mu] \left| \delta \vec{H} \right|^2 - \Re[\varepsilon] \left| \delta \vec{E} \right|^2 \right) dv = 0 \\ \int_V \left(\Im[\mu] \left| \delta \vec{H} \right|^2 + \Im[\varepsilon] \left| \delta \vec{E} \right|^2 \right) dv = 0 \end{cases} \tag{1.7.5}$$

For disipative media, both $\Im[\varepsilon]$ and $\Im[\mu] < 0$, implying $\delta \vec{E} = \delta \vec{H} = 0$ everywhere inside V.

1.8 Uniqueness Theorem

For the surface integral in (1.7.4) to be zero, that is,

$$\oint_S \delta\vec{E} \times \delta\vec{H}^* \cdot d\vec{s} = \oint_S \hat{n} \times \delta\vec{E} \cdot \delta\vec{H}^* ds = \oint_S \delta\vec{H}^* \times \hat{n} \cdot \delta\vec{E} ds = 0 \tag{1.7.6}$$

The possible conditions are:

i. tangential $\vec{E}$ on S is specified, i.e., $\hat{n} \times \delta\vec{E} = 0$, or

ii. tangential $\vec{H}$ on S is specified, i.e., $\hat{n} \times \delta\vec{H} = 0$, or

iii. tangential $\vec{E}$ is specified over part of S, and tangential $\vec{H}$ is specified the rest of S.

1.7 Uniqueness Theorem

Uniqueness is established for general lossy media, with lossless fields treated as the zero-loss limit of lossy fields.

The principle extends to non-homogeneous media by applying it locally to small homogeneous regions. The method is limited to linear media.